

文字记录：新录音 4 2026年9月18日

录音主题：Onsite Audit Followup Discussion

录音时间：2026年9月18日（周五） 10:51 - 11:57（GMT+08）

智能纪要：[Onsite Audit Followup Discussion 2026年9月18日](#)

说话人 1 00:00:35

achieved here. but we also have some suggestions, so we will to share this with you. you don' t need to put anything here.because some hours, so we understand that you may think about people at the best and really give you option.OK, so we want to explain one by one and to ensure we' re on the same page.

说话人 1 00:01:00

Yes, it's, yeah, maybe just to add a lot of context, like it's been a lot of really good discussion last few days and there's a lot of good progress.See some follow UPS and some actions we want to do review and talk through just to make sure it's clear kind of our expectations are. And we think that there can be some improvements.

说话人 1 00:01:21

So we compiled our notes here last night.This morning, this is giving you kind of overview, the sense of 60 some odd individual items.Some of them may overlap slightly, but generally there are three items.We try to prioritize these in a way that was maybe helpful for, so.I think we say high priority of this means there's more urgency and timing, but also just follow up on the discussion to make sure they're completed.So some of maybe the more simple tasks we can have was no priority in the sense that we can tackle those.I think we need to go back.

说话人 1 00:02:23

Currently after that.So we need that probably additional column.We did want to say for the time we have left and be able to work through, prioritize, talking to the items and make sure we're clear on what our notes are and then maybe our ad like.

说话人 1 00:02:45

We're sorry it took so long to compile.We tried to use some of the AI tools to take all three of our notes and it was trying.So we're gonna try to provide some clarity, but like just the way the AI

tool extracted, display the information, some of that may still be a little bit unclear video. So as you're reviewing this after we send it and trying to develop an action plan against it, there's any questions or clarity needed or. Can you send me a picture of this or drawing of that or, you know, feel free and don't waste any time to put their question.

说话人 1 00:03:17

But then 2, as you're reviewing your own notes and you also several notes throughout this, if you're reviewing your notes and you don't see something listed, you should escalate that because it may fall through the cracks of AI tool or not filter out.

说话人 1 00:03:32

This is a set of also kind of categorize these based on categories. Let's go through that generally here. I think clean relations is one of the areas we probably have made the highest concern currently around your strategy generally. I think we try to talk through that yesterday. There's hope, maybe not clear on our ends managing that generally. This together out here developing that cleaning program. I was actually executed and documented. So I think there might be some whole sessions on that. So this is sort of touching on that. Maybe if you have things in place, this wasn't the areas. That's great. I think we have feedback last time we were here as well. I mean, I think that makes sense.

说话人 1 00:04:58

I think I would highly recommend we need to set up some specific discussion. These actually three years of some frequency started to do that. I think there's probably discussion turning whether we can get comfortable starting validation knowing.

说话人 1 00:05:33

Not to like add more meetings, everyone's calendar, like I think like specifically dedicated to just that topic will be very. Stuff, you know, from our end to be saying those would be helpful.

说话人 1 00:05:45

But then like also just generally about this, I know so far the way that we've been approaching VOA topics in discussion is like here are, you know, you're recording like at off meetings that we maybe have here or there and we maybe have two or three debate. But maybe consider more target focus time. Like the jpts usually get kind of absorbed with more process technical discussion, and we don't need a ton of time for like the VOA readiness. So it just actually consider how to.

说话人 1 00:06:20

Yeah, I know. It's a lot of things that. See. Johnson, appreciate it. Yeah, that's all right. Every day we see that's up there. This was just a general lot of typing feedback. I think this one is maybe supposed to pull out around this line sloping. There's least us some lines of the period of these slopes away that they're not supposed to train points over this, like there's probably a follow from there and to holistically. I think it's probably a thing for a lot. I've seen most of the ones out there with one of the ones very carefully, obviously highlighted some examples of some of these gaps or opportunities, but I don't mean to capture everything.

说话人 1 00:07:42

This is, I think, another high level item that we did talk about yesterday. We were visiting your route for bringing materials in the building, have concerns around the lack of control around management of that material flow. So there's this place, the door to the outside directly in the C C space. That will be offered or airwalk, so there's probably some risk of contamination risk in this place that would be evaluated.

说话人 1 00:08:17

I think long term, that's probably the right thing to do for the operator or system. And then I'm feasible, obviously, the next year. So we can talk about short term, what. That's probably one of the higher status as well versus the left challenging team.

说话人 1 00:08:46

Think about that, what we can come up with for term and long term as well. I think your new design, you share this be a better design for new building. To have an airline with the pressure. And since that's coming up for the design, I'm going to be able to typically see, I think that clear. Any questions on that one?

说话人 1 00:09:30

There's a lot of those management discussions, I think there are a lot of discussions and feedback and good dialogue that's really helpful. Some direct recommendations. I think one is just moving this new board, trying to design. Oh, this is about that. Yeah, some examples we recommend that list of. So we have to upgrade those, you know, some kind of better, you know, we say there. Oh, fantastic. So that's what this is about. We should probably, yeah, maybe just to save our mental. Yeah, I just want to grab those, man. So we've got schedules. Catch all of them. Say I'm on top of it. So the other thing on those is that was just do ask, you look at the covers and there's the kind of.

说话人 1 00:10:28

Yeah, I was able to find the paper that we use and it's called Bio Shield. It's like the name of it. Say hi. Alright, that's enough to resolve. And then that was just, I think, a general guidance to me, like your characters change reasons, having some of the triggering that they're fine just to the, yeah, this was just the reason I had, it was fresh in my mind. Like there's a recent model time. At the park around. So there's just, yeah, a lot of people have just like literally painted a black line like underneath the whole storage rack, about 6 inches off the floor. Sometimes the hose is on the storage rack. You can see to the ends fall beneath the black line, just sort of check to check. And that nation from people walking and taking up things. So the back there with those seven different table covers. Justify it.

说话人 1 00:11:32

So the next, a lot of discussion, as I discussed around the period investigations, mostly around you got seven, some maybe also back in water. There's a lot of actions here. There's capture, maybe not exactly all these detail. Necessary questions we have. We talk about some potential gaps. We'll say time. It will skip through most of these, most of them that are clear at this point, touch points. So I think I'm not from why home, but. So this just like a caption that we talk about this. I think, but I will talk about passivation briefly to the other topic. So I do think the large team members in back of trial session last night start hearing there's probably concerned, more concerned on that team's project, the potential impact of lines to solve. Thanks. And others. So yeah, probably worth talking that group that seems there and driving. That's it. We can understand that.

说话人 1 00:13:30

Yeah, we're trying to be honest, but let's be honest. Yeah, sure. I think the night, sure, the stories have and maybe a question that typically our vendors have to be because that's always see. It's wondering that's common here in China or there was a gap, there's a.

说话人 1 00:15:34

This is also related in pure investigation. Just want to understand how, if you got. Or you get your documentation controls to ensure you want to like sort of setting 1 and that's it.

说话人 1 00:16:31

So I think in general, we had a great discussion on this. You have some procedural guidance and corporate guidance on the tradition strategy. Documentary for the strategy is you can find some procedures justifying filters you choose to use, type of occasion CCS process, but that's kind of,

this is kind of placement criteria. So. The last one, I think we put as a high risk, just primarily because I think we usually weren't planning to filter with the raw materials, with the rescue gas. That one, that's really expectation. All that stuff, there's this crush on filtration size and plan and choice of filter.

说话人 1 00:17:32

So again, we tied back to some of the things time back to your strategy and be consistent. But as you can see, all the stuff that you use to document that. Yeah, that's a great biggest point. So I'll try to share it in general. Anything that is specifically stated in that, you know, strategy document that was sent, that those are items are the highest screen because that's like. Approve by the Lily Speed Panel as like a global mandate for us to follow. And like the liquid charging of drop stock materials or raw materials is explicitly stated in there.

说话人 1 00:18:14

So if we don't do it, we have that really strong discussion not to. So, and I also have been with it, but like on a risk based approach to the filter placement, like again, just considering that like for a non robotic solution, like a mobile phase, do like you really need to do that every batch or are you actually putting the process at a higher risk by opening the system to replace the filter. Then you would be just like lead that for say three months, six months, some time based duration instead of following the one time use policy across the board. So I think those are the two biggest coaching.

说话人 1 00:19:06

I want to be very careful on like. But you guys have internal guidance and procedures that you're talking to. So like, I appreciate the evaluation, but I want to make sure that we're not going to get you guys in trouble for going against like that document you want yesterday made stated that this is what you guys do. So I just want to make sure like you guys also have that liberty to like maybe take a recognition, you know.

说话人 1 00:19:55

Next stop is just the love that's raised around the dispensing area. Love that's like the discussion there. One, you know, came out of just wasn't recommending all the likes to press it.

说话人 1 00:20:23

Talk about a lot about cross contamination risk, I think in particular around risk of multiple product facility, you may have multiple connections to your tanks that require your process so that we can assess them around evaluating those. There's, if they require.

说话人 1 00:21:04

Risk assessment, the valuation and the document and the approach. So our goal, standard expectation is if it's not required to run the process, that it's physically disconnected and locked. I was, the management touched on, G P touched on. So I think we talked about this one in some detail. I was interested to that's up. So yeah, I think. So I didn't talk to this one in detail, but like, I think again, like the gold standard would be basically on category 4 equipment. So broken solutions and product contact.

说话人 1 00:22:17

We typically wouldn't install a PD pump cuz even some of those specialized PD pumps, they still have internal checks or mechanisms preventing backflow through them for efficiency purposes, but that backflow prevention means that you can't get full drain ability. So I imagine a diaphragm pump, even if you have a low point drain right on the outlet, there's still gonna be some vertical portion of the outlet manifold and a little bit of piping that isn't gonna be able to be drained into that low point.

说话人 1 00:22:48

So that's why I think medium risk is a long term. Like long term, we could look, is there any like remove the use of those on growth promoting or product contact lines through either a just the use of simple pressure transfers? Like, do we really need a pump to provide the motive force? Or do we have managed to headspace pressure that is adequate enough to do some transfer? Or b, if you do need a pump for the force, you can go to a center fuel pump, hygienic center fuel pump, which goes that you Casey vote going on. There's no check valves are fully drainable. And then you can either use a VFB on those to modify the speed or control that on the outlet if you need to use it for metering purposes or control purposes. So. But in the short term, probably, we're starting to sense like the reality is more of those are gonna exist. We talked about before making absolutely certain that every one of those pumps has a low point on the inlet and the outlet, so we can drain as much as possible. And then following those draining activities that we make sure all the operational documents. Booking out the low price. Nitrogen source and kind of blowing nitrogen merging through the pump to let those checks lift and cycle to the low point during the outlet so that we can at least show that we made good faith effort to remove any hold up and leave behind as a minimum residual. So that's kind of the first high risk point there, just making sure that all our operational documents and clean documents have those structures built in.

说话人 1 00:24:25

The second point is the other downside of the pumps is they're usually like take a diagram, for example, a large surface area, soft component. So a lot of times clean studies like swab recovery factors, rinse recovery factors, they're usually done like stainless steel because those are the, it's all or whatever. Metallic LC is, and sometimes they may not be represented on these large surface areas of parts. Just making sure that your cleaning strategy accounts for the use and are we have a swab point around there, and we assess for every factor off of the soft parts, etc. So I think that's not the ones I think we're gonna have to be aware. We're gonna run with them, but just making sure we do everything. Within the realm of reasonable, this start to make it as fast as possible.

说话人 1 00:25:23

Yeah, big one. But so, okay, so, but at first we're trying to type up the list and we're trying to be very prescriptive on every non compliant thing that we saw there. But again, we didn't have time to walk every single line painstakingly. And again, it's like, I think we just need to talk to the theory and then you guys go to liver.

说话人 1 00:25:45

I'll say before I came on site, based on some of the Excel spreadsheets that were shared, flagging what gaps still existed on the equipment, I was expecting to see less gaps out of the floor. But I think when I look at specifically the Category 4 equipment, which I care about the most, but I care about all that media category, but. The category 4 points of growth, rolling apes, base solutions, water and product contact slash clean contact equipment. But again, the expectation would be no ball valves, no flanges, no hard to clean areas, etc. I think we saw a lot of themes. I think some of those themes, this is here. So again, ball valves. Like we saw ball valve on the top of the E, the purified water tank T01, and ball valves all along its supply line, etc. Some examples, long tank nozzles on vessel heads. So again, we talked through other pictures, kind of the ASM standard on what a reasonable nozzle this. Sometimes we have very long nozzles that we don't believe are gonna be covered by sprayers, just too much distance up there. But then 2, sometimes it would get worse for the long nozzle. And on top of that, you would have a T for some other instrument. So now you have like a branch that's certainly not covered, you know. So about all those dip tubes supported by hooks instead of just like a straight tube valve so that the hook where it grabs the tube creates a crevice along that things can get up on missing drains at low points. So like a lot, this would be like specifically around, it's all off. Filters where you're not gonna get back through the filter element, like just you're not gonna pressure oftentimes to push it and we would have a low point in the filter inlet or outlet on the other. So again, just a key pumping water low point on either side of it. And even if there was a low point, it was often like one of those little valves, very small. A diameter where sometimes, just like surface tension alone, drain ability through those can be difficult for us to like purge through it. So assessing low point drains should be some of the instruments definitely are not hygienic and cleanable design

and don't need to kind of course, two biggest ones I saw are those. Level transmitters, where there's just a horizontal, two horizontal ports coming outside the tank with a paddle, GP paddle on each one. So there's just gonna be a stagnant horizontal liquid sitting there that, you know, when you drain the tank, you're gonna make a puddle on that. So again, we would typically do like non contact radar on the tank where it's just. I had the vessel and it's just a small probe that comes down and measures that never in contact with the liquid. And if we had to use DP or wanted to use DP for some reason, they make those flush mouth hygienic panels for you actually integrate right into the.

说话人 1 00:29:00

Bottom of the tank. So when you're averaging the vessel, you leave a hole in the bottom and actually weld the panel directly to the tank so that it's just like continuous and smooth with the surface of the tank. Go over there and then the other one usually goes on the other side so that, again, you're going to the other one that we had to confirm but I'm worried about is again for temperature, I used to see a small thermal level right in the dish of the tank. You know, and it's all just, it's just a seamless tone, essentially. And the RTE just slips in the thermal. The RTE actually never touches the product. And the thermal itself is smooth, cleanable, and it should tank. I think where I saw a lot of these vessels is the temperatures at the top of the tank and requires a super long bit tube to bring it down into solution. And I guess it'd be better, it's also that sealed thermal design. Maybe I can get there. It's an open dip tube that like as you fill the tank, that inside surfaces are gonna fill with your product and then you drain, it will empty, then you're spraying only outside the dip tube. You have no way to clean the inside of it. So that was another one.

说话人 1 00:30:09

Flow meters, several of them were installed in the non drainable orientations. There's horizontal with the belly or the low points low. So best case scenario in the vertical or possible and if not, at least rotate them 90 degrees so that it with the slope of the piping, it can back work its way through those tubes.

说话人 1 00:30:31

Flange connections on category 3, category 4 equipment. Still several bolted agitator blades. So first we thought it was just our five and we found it in some of the diso tank ones. And then we didn't need to go look at all the tanks. So I think just a full validation on those would be helpful. Dead legs in various locations do. So basically not using the non business teas or zero static valves. So like some obvious ones we saw were like bottom of the precipitation tank that go out, those drain valves. Like you have, you come out of your tank, you have your product flow path coming out of one leg of t, but on the bottom leg of t just be a dead leg going to drain

valve. That presentation was one example. We saw it almost everywhere where that was, again, zero static or medium distance and you wouldn't have that there. The other place I saw it that wasn't on drain valve like concerning it was like fractions and outlets where you have this, you come up on the tank and you have two paths and you go, here you go. Waste or to mainstream. Those both have long t's on them. So let's say fraction 1 of injection 1, let's say that's going to waste. Now you fill that whole bottom animal, including those effects waste material and you know when you go to change that, you're gonna leave a pocket there. And then now you're gonna go around your second junction, let's say, for whatever reason that qualifies for Main Street. Now you're gonna push a slug of that waste material into your main street. So yeah, on that topic, I just wanted to do my team do your estimation about that line that underneath the fraction time that you saw, is that the. You know, the future life on that highlight.

说话人 1 00:32:21

Remember that we saw the live photos? Yeah, I told him it was a filter. And I think the last time I was removed after seeing you. Okay. Yeah, I'm also like a notification for this, like I can. Yeah, and like, again, that's where, like, if you want to draw up kind of what you're thinking to handle these drain valves and, you know, and send me a picture of the sketch or something, you know, I think that'd be helpful cuz then I can say, yeah, so those are kind of some examples. So basically, I think the last stage here is like. It's just gonna take someone going painstakingly through the equipment and like just walking every single line down and saying, if I have a broking solution, a product containing solution or cleaning solution, if I run across a ball valve, I need to replace it. If I run across a flange, I need to replace it. If I run across a dead leg. You need to address and it's just gonna take someone doing that. Very easy for us to align.

说话人 1 00:33:32

And then ideally, you know, the different sheet I did, but like ideally, you know, we were meeting as many of those as possible. And the ones that we actually can't do the lead time or resourcing, whatever we need, right, we're gonna need to make sure we have 100% traceability to those so that we're more reviewing clean documents, etc. We need to make sure. Or at least blowing through those and cleaning those, etc. Right.

说话人 1 00:34:01

So like maybe give it, for example, give it a zero static drain valve. You wouldn't have to toggle that drain valve very clean because it's gonna be clean just during no safety. If you can't replace it with zero static, you're gonna be getting down the drain during the clean cycle. Someone's gonna have to go out there and toggle flush clean solution.

说话人 1 00:34:20

Through the line, you know, so was, but we need 100% traceability in order to be able to make sure that was a big one. Yeah, this is probably one of the most challenging. I think it's gonna take a lot of efforts and a lot of dialogue and discussion to make sure. One year, you're able to kind of walk through, systematically address these, see also some of these apply like, yes, yes. So it's not just crazy, although that's what high risk is, having that legs and marking mazes and actually. And I know that we shared like the different categories and the flow chart on when they apply, but if there's any confusion on, hey, does this need to be category 3,4,2,1, right? Just email us as we can clarify it and it's gonna be on my line. So basically, this is your whole category. So the other thing is, just to clarify, is that these are high for those, for example, water base, solution base, that's like you as the night trial, you saw that it's okay. Say, say, and then most of the time, say for example, this like 2% estimate, right, or above is okay, right? Is this something to do, right? And then something below, say, percent estimate, right, to the water base that I apply to all the equipment. Yeah, but I'll say that the star. So if you read the document that was trying that, right? It would say that like as best practice, even on those non growing buffer only systems, you shouldn't have that low points. You can't do anything because of the fact that you guys are a multi product, multi client system or like, let's say today it's ACN, but tomorrow it needs to be something else.

说话人 1 00:36:33

But I think from a risk based approach, like to start up, yeah, you're right. It's not more money than we can, which is consider times. Yeah, exactly. Basically, we we won't we should clear label that this is that, you know, capable of people. Exactly. That's okay for people. So that's more, you know, anything now that we have. Yeah. And then, but longer check that we should do that we should get all changed, which is just considering timescale before. Yeah, yeah, yeah. Yeah, but I agree with this sense that when you're going to evaluation, it's not products, it's the buffer. And that buffer is not going. So it's organic in nature that I agree. We ball valves, you can be low points, you can be big nozzle or ball nozzles, but the ones that are in scope, our product, yeah, and the product, it doesn't matter it's stroke or not, it's product. It's it. The scope growth, promoting alias buffers.

说话人 1 00:38:19

So like example there, we talk about specifically is like, it sounds like right at the top of the day, we still know it a couple flange connections, but it sounds like that one might be some hurdles to getting done quickly because of some pressure vessel certification hurdles. So if that's one that we have to leave there, then like I would expect that when I pull the clean up in, I see name clean instructions for it, like inspections, etc. You're right. Okay. So for example, some of the long numbers are we say, trying to get it more CFP, but you maybe, for example, you can check out. So then, so hard actually at this then.

说话人 1 00:39:15

Yeah, I can visualize you guys on installing that, taking this relatively small valve to some. Clean car, clean and point it. I can visualize that. But like, let's say you have a super long nozzle with a temperature pressure element on top there. Like, are you really gonna like master the calibrated instrument and hold that off and potentially impact the calibrated state just and then even if you did do that, now you have this exposed nozzle. How are you actually gonna. Clean it. Like, it's not like you can take it off the tank and dump it. So you don't like, so that maybe I would just talk about my like face a lot of just underneath those problems, right? So we need to be okay, for example, we say we splash. Yes, that's possible. But we should have some sort of support. So that's it, right? Yeah, because our requirement is that there will be possible those scenarios, right? Exactly. So our requirements that we create, yeah, and I will never learn that for longer time, you know, we will replace. Yeah. It's just consider short. Yeah, as I've got it.

说话人 1 00:40:41

Yeah, I think the first big challenge is to define, yeah, what we have. Yeah, it's a big test. Yeah, there's a concern. Just that's a big challenge. That's right. So you guys can give us both. This person was at the. And I know that we try to sort of do some virtual collaboration, like on this topic where I saw some piano here, but like sometimes I think that was difficult because I mean piano doesn't, yeah, we do outside, we do conference with camera outside. It has a meeting. And we should say maybe that we label those, you know, you guys have to say this is it.

说话人 1 00:41:28

And it could be live on that. Yeah, collaborative or like, honestly, those videos that you guys shared up. Okay, share videos. Exactly. Like that's where we picked up, like watching a piece of video, that's where we picked up a plan. So he just had someone create a video where, I mean, those were like showing sweet. So they were a little higher level, but you didn't. In specific videos in this. And someone literally just looked at every connection on top, every connection on the bottom. We need to watch those and flag. There's this. So whatever you find is most.

说话人 1 00:42:03

Okay, great. Thank you. Sorry, that was a big one. This is the other large one. And we talked a lot about this. So like, again, purified water, the fact that like our sample point is over here and then we have these super long headers so we don't have actual sample data at the point of use. So I think at a minimum, what we describe, and then sorry on that one too, when we have these large headers. There are sometimes they had multiple client tanks, but we had no isolation on each individual one. So let's say you were trying to feed water to ATO 4 to get ATO 4, it passed a header and there was all these big deadlights down to all the clients. Those will fill with water

and you have no way of getting it out right now. So I think at a minimum, we said it is. The headers need to be redesigned to have isolation valves at each client, so if you're feeding a client, it has a direct path that's not gonna fill any other deadlights. And then we also said at each client, we need a drain valve where you can do a pre use flush to waste out the drain valve before you supply the tank.

说话人 1 00:43:21

Now, what would be great is like at that new drain valve. That is your point of use, that's what's representative of your tank. That's where the purified water validation sample point should be so that if anyone asks for this tank, the water supply, how do I know this is meeting quality? You can show them from this point, right on top of the tank here with my sample tools, right? So I acknowledge that it's unlikely to maybe revalidate the system following modification. So I think at a minimum, we can rework the pattern piping and. Give ourselves a dream path and maybe provide some justification like here's the nearest sample point and here's these other controls we have in place. Maybe we can get there or maybe we do have time if like a station source or something else where we could put it in and do a validation on those new sample points. Or maybe there's an in the middle approach where maybe we don't do a formal validation, but we at least pull some grab samples from each and do an entry sampling or something to give us some more confidence until we can do a full validation. So there's something to consider there.

说话人 1 00:44:36

But again, I think you guys understand the minimum requirement. Get rid of all the links on the water system. Give yourself a way to do. Reuse, flush and give yourself a way to, when you're done using it and that whole line is just still sitting, for water to blow it to that new drain and empty it out.

说话人 1 00:44:56

And that was another one of my big ones is, and then I shared the highest PE guidance. So if you pull that up and read, that walks through these very specific examples of what they expect.

说话人 1 00:45:15

And then just, and then a reminder like this, No. 94. So like this, I was when we were looking at our 0,5, which is a growth promoting solution, there was no point of use water filter on that line. So. I think our practice would be filters on water lines inherently can have some risk associated with them. So we don't want to use them if we're going into a non recording solution. It's not worth the risk. But when we are going into a growth growing solution like that, our 05 time we want to point use over there because you're gonna hook up this hose that's been sitting in a

CNC space. And I always want that filter to try to catch whatever we can. So I kind of how reserve like that as an example where I didn't see a point of use water filter. So that's kind of one piece of it. The other flip side that I guess is like on the water supply to like 84 or 5 or 6, that one there wasn't filter there because it's coming into a pretty deep space. But then I think we were unclear on like how long can that water filter sit there without being replaced? Because to me it'd be like, you don't want to go any longer than, say, 24 hours without replacing that water. So the memory itself can't grow anything because even if you pull the line dry and you don't put it, it's gonna stay wet.

说话人 1 00:46:43

But I think that's another part of our system. But I think that was like another, but I think honestly, there's two things out of all trip that I'm most worried about is the one we talk about the gaps on category 4 and 3 equipment and this one, those are the two biggest topics that I see quite a bit of work. And the question, so for the. We're talking about the sampling points, like on the times, right? So it's the expectation to take sample like before we use each director, like each time. Oh no, what's the like, so like, let's say we have all the time in the world, right? You go. Or maybe, I mean, and then you would go do, I don't know if it's like a three phase validation on your water system we usually do, or is that how you guys validate your water systems out? Usually so I can install your water loop. You have to put your validation, right? And in the States, it's a three phase approach where you're each phase is a different length and you're monitoring like daily during phase 1 and phase 2. But then you're monitoring weekly or monthly in Phase 3, and you're checking that each sample point meets the purified water specification or GMC, endotoxin connectivity, GOC nitrate, etc. So ideally we'd all time around before we ever made any medicine there you read, I mean, go put the system through a phase 1 and phase 2 validation. Show that each of those sample points passes. And then during routine production, you would just do what you do on every other sample point where it's what monthly, my guess is how much you sample. So you don't have to sample for every single use. You just have to validate use point and sample.

说话人 1 00:48:31

For the short term, I was recommending is if you can't get a whole validation done for these new ones. Maybe at least like whole 3 days worth of data or weeks worth of data, some abbreviated validation to give us confidence until we can go do the longer validation or maybe do like a concurrent validation sort of thing. So just something to consider. Yeah, this is really smart. Or no, this is a little bit, this is just, I think, a request that I made by maybe one of the super creators on platforms. This was like. For any equipment that is not brand new but being used in the process, having a list of that equipment and then having a clear changeover approach to understand how we have full confidence that whatever product was in there before is removed before our product. So that's what this one is like. And I think like at the surface level, you would just say,

well, we have cleaning limits for it. For the change over, we're gonna run a cleaning cycle or spot prints. I just gave some examples of like things beyond that simple surface level that like maybe you're often missed like that lines for example, that could have seen product a, but you don't go disassemble that line and then we clean it. Or like soft parts like diaphragms, hard to clean components like filters, TFF membranes, repacking column resins, kind of a holistic change over approach for anything that's being used. It's brand new. We don't have answers.

说话人 1 00:50:33

Yeah, this is very similar. This would be for brand new equipment that has a different risk profile. Now your risk is, hey, I'm carrying over this whole product. Your risk is, did I leave something behind during fabrication or construction?

说话人 1 00:50:48

You know, so like, I mean, you guys know exactly. Yes. Use black collar involvement. How do you know that's removed? When they weld and pop? Do they deposit metal particulate into the tank when they weld? They put those like purge plugs in the purge gas. We've seen those purge plugs in lines, you know, just like there's a whole slew of things that can be left behind from.

说话人 1 00:51:14

Construction, fabrication effective. So what specifically you guys do to confirm that before those into this first product use, all those construction related debris are removed? So I think we talked about some of the elements in the conference room yesterday, like internally, for example, we do like a white test. After we clean the vessel, we go take like a solid saturated white. And white, various regions of the day, sidewall, each quadrant film, each quadrant ditch each quadrant, right and show they all come up perfectly white still that you don't have any polishing residue or metal particulate. We would then screen or a filter on the outlet and do a CIP of the vessel and collect everything out, and then. Pull the screen out afterwards and make sure there's no degree collected, right? So there's layers to the strategy to say, yes, this is cool.

说话人 1 00:52:05

Yeah, I believe what they're saying is we can do the person that could have something like that. Exactly. So yeah, I think this will hopefully just to provide us what your strategy is and we can look at it and say, yeah, yeah. You know, we kind of touch on this thing. There's still a few questions there. Session, maybe a bigger question. Yeah, it's just confirmation you have like the boundary of your coverage and if there are components or parts analysis to be tested here. Question generally and did a few questions on the strategy generally community that's really 18 see in those meetings and discuss there.

说话人 1 00:52:55

It's like, I don't like internally very many guys mention like you look at pressure. And then maybe, let's say during your recovery test, you show we need 25 pounds of pressure to get your vessel clean. How does that translate to when we're actually running clean cycle? Do they have a pressure alarm or something to let them know they drop below that pressure so that they might have to be covered. So that's 1,2. I think our cleaning scale will have this thing look like. We've gone away from pressure and we go to flow, specifically because we saw instances where like if you have debris or particulate get into your system, they can partially block the spray balls. So your supply pressure can be the same. You're taking more pressure drop across the spray lines, so your flow is lower or you're partially obstructed. So like he might have some feedback on that. But I think this is just, they discuss a little bit more. I think our groups need to be struggle. So it's more than I expected. So I was wrong. I mean, if you're looking at it, we're using more solvents or lower generation than this. If so, we may still have some buffer based on these Assumption. Yes, we can say that it basically leads to a higher rate of, of this, more of the volume of this problem that we require. For the case of yesterday, you can hire if we using 10 times the page and we all pass, which means we must using more solidly. And motivation. So yeah, let's check it. We can be even change.

说话人 1 00:54:57

Oh yeah, I think you guys are well in progress on this one, but it just remains the highest Mark is. Several places drop process. We use a temporary vessel to store and transport water. So some examples unify our whippy t f F collaboration water. And if they put on the right investigation, s t d s water or. So we have these, making sure that we're doing old time studies on those to show that, hey, I filled up this container on day 0. How much time do I have where it's still needing my specification before I can drain it, clean it, refill it. So I think those are in progress, but just putting it around here.

说话人 1 00:55:47

Cuz again, like best practice would be wherever you need water, you have a point of use direct to it. You don't have to put it into something. So what are those there? So charging spaces, I think our structure technically requiring heating, charging of all sounds. So just, I know you have the capability generally to put our house. That's probably the best transition that, you know, all sorts of charges possible. So you could understand like, yes, yes, and leadership that's coming first, that's feasible if not in the success short term and long term approaches there. Yeah, definitely product quality concerns, more environment concern, especially given that these are multi product environments.

说话人 1 00:56:58

So like the example of the situation that we used to say was R05. To charge the trust base. That's just gonna be a zip tie with an open try frame on a normal butterfly valve. So after you do that charging, you're gonna go disconnected and any residual in the shoe has a potential. And then you're gonna have the top side of the butterfly valve that's gonna have power just on it. So, and then that's a CNC space where you could be running other products and. It's where the operators don't have any overguard, so that any exposure can land on the ground, which then could have some risk to classified spaces if they bring containing down into those airlocks. So that's why I think the strategy, you know, it seems a little unnecessary. That's why the strategy is here to says.

说话人 1 00:58:17

As you're walking down the lines, even if you see a diagram now, make sure it's installed at the right level. Again, this is just a, once we get press highlights, just several more summary. Perfect. So yeah, let you guys kind of send this over as soon as you know there's questions, there's that clear that stuff, and then you may update this and then there's probably a little more. I'm working direction on advertising. Like we're gonna get a chance to have this conversation today.

说话人 1 00:59:00

Yeah, yeah, yeah, ladies. Each focus on several of these topics. Yeah, some of these are probably much larger, I think, and the timing is one.

说话人 1 01:00:04

So, we will provide the business special. And then I think the scene can help. Now, when they're working together, there's a one meeting. So. Also the discuss like if necessary, like another outside visit down the road before you can start again. It's just really close. Like I have a lot of people rotate. So maybe some of them just kind of look at stuff like if any video ideas like the video, like, yeah, that might be good.

说话人 1 01:01:28

But again, like, I think if there's anything where you just want to put out before you go and invest a bunch of time and resource reworking it, like. Okay, here's what we're doing for training now. So you draw up and send it to us. We review it, say yes, or, you know, hey, the fraction chains, here's how many early deadlines do you want this design? Anything you put in front of us before you put it in front of us. Exactly. First. Okay, great. It's not so much, yeah, to show me to use that time when we have a quick video and then you guys have a lunch, right? We're gonna

be, yeah, after, yeah, six minutes like this will probably come easier. Six minutes, I'd say. You're gonna probably share stuff. I can imagine the broader releasing so I can catch up with all of something five. So Peter's, yeah, the, I'm gonna share this draft version. We'll talk about. Yeah, right.

说话人 1 01:03:36

There is our general requirements for our management of the task, including the general requirement such as the video for each type of software, fresh software, including the task and yeah, and all. And before use, we will perform the qualification of the test, including the IQ and the PQ.

说话人 1 01:04:05

In different qualification, we have several different for the, such as the cleaning or the test and the cleaning of the security cabinets after the IQ and the PQ, we clean the pipeline and test. We will also confirm the testing for the for the solve on the animals such as clean after test, we will charging the first solving to the test or release and then this charge from the bottom value of the test. There was no parameter or error. Every parameters and if another is the each headline from the time from to our production workshop and we will use the we will take sample from the equipment to confirm the something gives the price on this specification as well as our we and here is our positions and as last is our.

说话人 1 01:05:11

And you're holding assessment, including the specific issues of the first order and the stuff in case of the impact that has during storage, during long term storage, we can, we will perform the every test, our test every year. So here.